Options data-source and archive audit

Generated 2026-08-14 02:23

> *Options data-source and archive audit for Phase III. Archive audited on 2026-08-14. This is a plain-language explainer, not a technical reference -- it describes what the archive is, which parts of it Phase III actually uses, and what the known gaps mean.*

What we set out to answer

Phase III of this project asks whether the post-earnings-announcement drift phenomenon that Phase I established in equities also shows up in options prices. To answer that, the phase needs option-level data (daily prices, bid/ask, implied volatility, and Greeks) matched to the underlying stock and to earnings-announcement events.

There are two possible sources of options data for this project: (a) OptionMetrics via WRDS -- a modern, subscription-based feed running through 2024, and (b) a local OptionMetrics archive from a prior collaborator, covering 1996-2013, on an external drive. This report explains which one we're using and why, and what the archive we chose actually contains.

The WRDS check outcome

Result: OptionMetrics is not available on this project's WRDS account. The account's list_libraries() call, saved to data/metadata/wrds_available_libraries.txt, returned only 'optionmsamp_us' -- the free 30-day sample database, not the full 'optionm' schema.

Consequence: Phase III uses the local 1996-2013 archive at data/raw_options/OptionMetrics (symlinked to /Volumes/copyrishabh/OptionMetrics). All the H1/H2/H4 results in this phase will describe options-market behavior in that window, not the post-2015 period. The 2015+ WRDS window (called 'Path A' in the plan) is closed by unavailability, not by choice.

What options data actually is

An option contract gives its holder the right (but not the obligation) to buy (a call) or sell (a put) 100 shares of a specific stock, at a specific price (the strike), on or before a specific date (the expiration). Every day, dozens or hundreds of contracts trade on the same underlying stock at different strikes and expirations, each with its own price.

For each contract, on each trading day, the archive stores:
- a bid (the highest price a buyer was willing to pay at the close),
- an ask (the lowest price a seller was willing to accept),
- a volume (how many contracts changed hands that day),

- an open interest (how many contracts existed at the end of that day),
- an implied volatility (the volatility number the market implicitly used to price this contract, backed out of the option price),
- Greeks -- delta, gamma, vega, theta -- which measure how the price responds to changes in the stock price, in market-wide volatility, and in time.

The multiplier -- normally 100 -- says how many underlying shares one contract represents. All of Phase III's later work assumes this vocabulary.

Summary of the archive

Archive audit ran against /Volumes/copyrishabh/OptionMetrics. 102 SAS source files, 65 parquet files converted, 37 SAS files without a parquet twin. 0 error(s) and 1 warning(s) touch tables Phase III actually needs; the rest of the 37 finding(s) are on tables Phase III does not use.

What the archive contains, table by table

Every row below is a 'table family' -- for the yearly tables (opprcd, secprd, hvold), there is one parquet file per year in the range shown. 'sas_gaps' is the number of SAS source files that were never converted to parquet (either because the SAS file was found corrupt or the conversion job was never run on it).

family parquet_files sas_gaps year_range relevance description
---------- ---------- -------- ---------- ----------
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distrd 1 0 non-yearly required for Phase III Distributions: splits, cash dividends, spinoffs.
opinfd 1 0 non-yearly required for Phase III Option contract metadata: strike, expiration, style, multiplier.
opprcd 17 1 1996-2013 required for Phase III Daily option prices (bid, ask, IV, Greeks) per contract.
secnmd 1 0 non-yearly required for Phase III Historical security names / identifiers.
secprd 19 0 1996-2013 required for Phase III Daily underlying-security prices.
securd 1 0 non-yearly required for Phase III Security universe (one row per security).
zerocd 1 0 non-yearly required for Phase III Zero-coupon risk-free rate curve (input to option pricing).
stdopd 0 18 non-yearly not needed for Phase III Standardized-tenor synthetic options (constant DTE/moneyness).
vsurfd 0 18 non-yearly not needed for Phase III Volatility surface at standardized DTE/ moneyness grid.
exchgd 1 0 non-yearly informational Exchange codes and calendar.
hvold 18 0 1996-2013 informational Historical realized-volatility statistics per security.
idxdvd 1 0 non-yearly informational Index dividend distributions.
indexd 1 0 non-yearly informational Index metadata.
optionmnames 1 0 non-yearly informational Historical mapping between option IDs and their standardized names.

opvold 1 0 non-yearly informational Aggregated option volume and open interest per security-date.
securd1 1 0 non-yearly informational Security universe variant (alternate cut of securd).

Findings that matter for Phase III

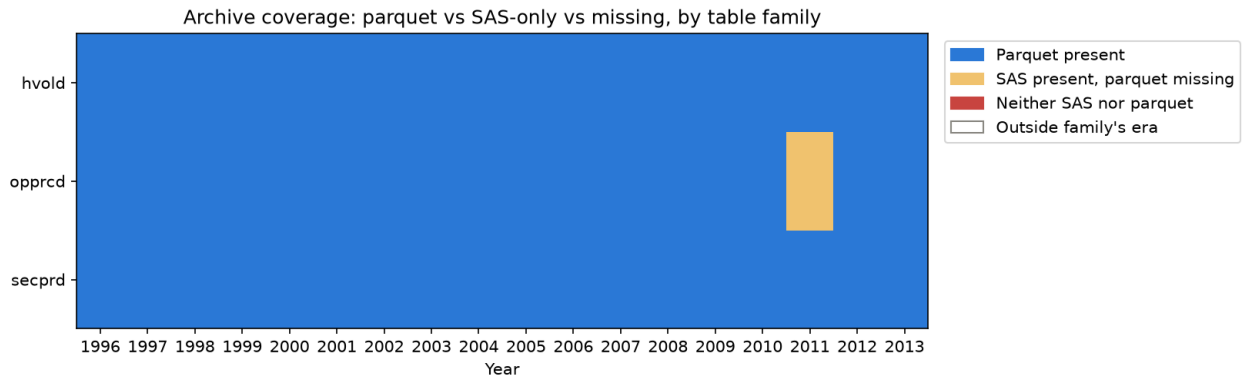
Warnings (documented failures, may be recoverable):
- SAS file opprcd2011.sas7bdat has no parquet conversion. Reason: [11:02:00] CORRUPT/UNREADABLE opprcd2011.sas7bdat: A row in the file was not the expected length — skipping

Other findings (catalogued, not blocking)

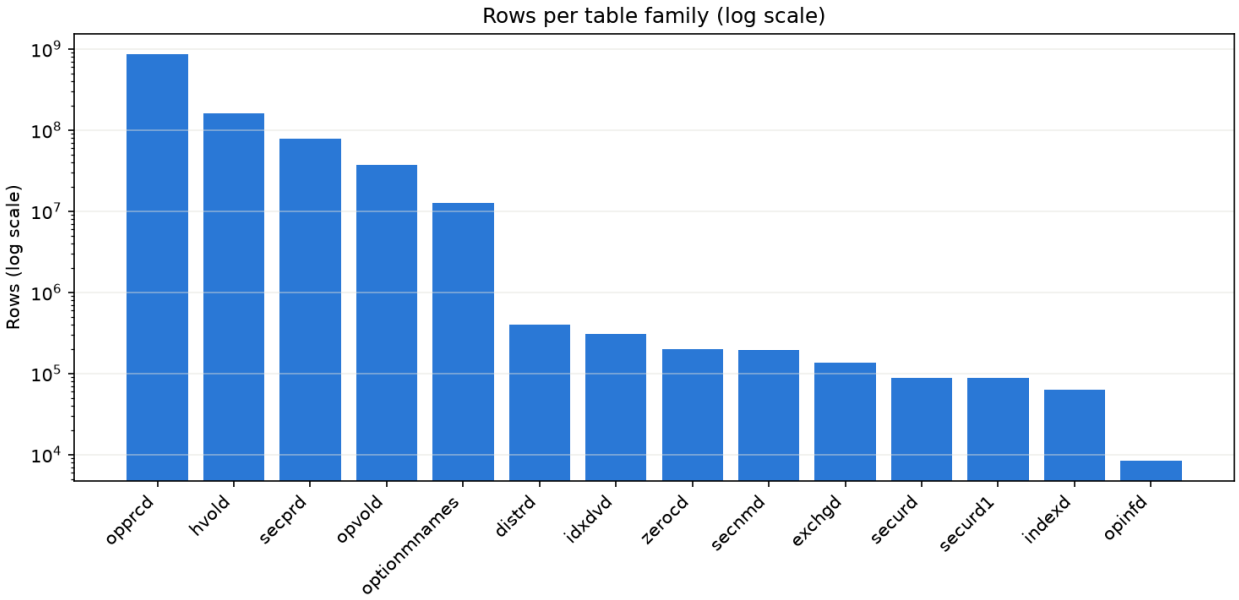
36 error(s) and 0 warning(s) touch tables outside Phase III's requirement list.
Affected families: stdopd, vsurfd. These are catalogued in data/metadata/options_archive_gaps.csv and options_archive_summary.json for future reference; they do not block this phase.

What comes next

This report covers Slices 2 and 3 of the phase's plan. The next slices (4, 5b) define a canonical option-observation schema and a loader that maps the archive's IvyDB fields onto that schema. Slice 6 wires up the data-quality gate the plan requires, and Slice 7 does an independent Greeks/IV recomputation on a stratified sample using QuantLib as a cross-check on the vendor's numbers. None of the actual event-study results (H1, H2, H4) begin until Phase II ships the equity strategy work that they either build on or need to control for.



Archive coverage per table family, one row per family, one column per year. Blue means 'parquet file present.' Yellow means 'we have the raw SAS file but it never got converted to parquet.' Red means 'no SAS file, no parquet file -- a true coverage gap.' White means 'this family does not have a year at all -- it's a non-yearly table like opinfd (contract metadata) that lives in a single file.'



Rows per table family, log scale. The scale is logarithmic because the biggest table (opprcd -- daily option prices) has hundreds of millions of rows while several others (opinfd, exchgd) have tens of thousands. A linear scale would hide everything except opprcd.